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Glory of the Garden

Glory of the Garden 



Easy

Forensics

picoCTF 2019

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Description

This file contains more than it seems.
Get the flag from [garden.jpg](#).

Hints 

1

What is a hex editor?

91,685 users solved



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picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: Glory of the Garden

Category: Forensics

Date: 03/29/26



I. Objective

The objective of this challenge is to understand how files (especially images) store data internally and how hidden information can be discovered using **hex analysis**.

The goal is to inspect the raw data of an image file and locate the **embedded flag**.

II. Background

All digital files, including images, are made up of **binary data (0s and 1s)**. To make this readable for humans, this data is often represented in **hexadecimal (hex)** format.

In forensic analysis, investigators sometimes inspect the **raw file data** because:

- Hidden text may be embedded directly inside the file
- Not all data is visible when opening the image normally
- Some flags or secrets are stored in plain text inside the file

This challenge demonstrates how to **search inside raw file data** to uncover hidden information.

III. Tool Used

- **Hex Editor (online tool)** - Used to view the raw hexadecimal content of the file
- **CyberChef** - Used to decode hex values if needed

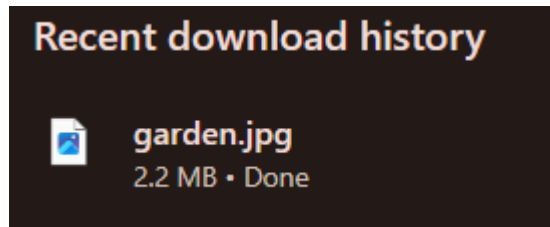
IV. Methodology



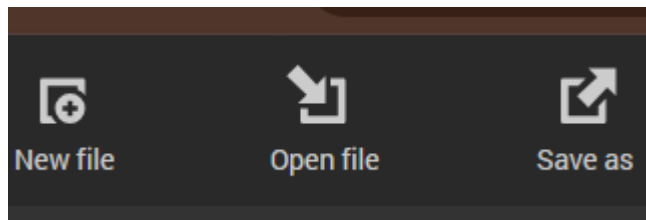
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1. Downloaded the image file from the challenge



2. Uploaded the file into an **online hex editor**



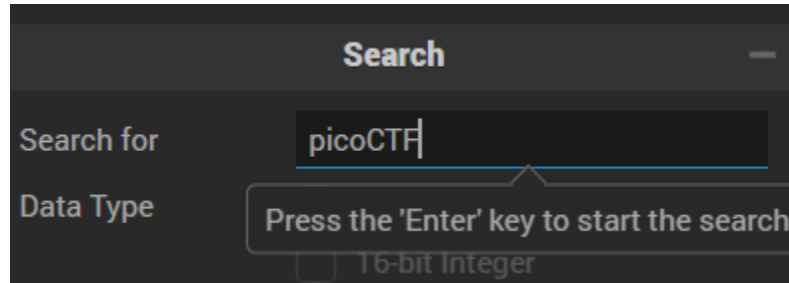
3. Viewed the raw hexadecimal data of the image

Untitled- x	garden.jpg x
00008C0	40 E7 41 29 41 6A 41 AC 41 EE 42 30 42 72 42 B5 @tA)A jA%AeB0BrB
00008D0	42 F7 43 3A 43 7D 43 C0 44 03 44 47 44 8A 44 CE B=C:C}C^D.DGDèD+
00008E0	45 12 45 55 45 9A 45 DE 46 22 46 67 46 AB 46 F0 E.EUEÜE F"FgF½F=
00008F0	47 35 47 7B 47 C0 48 05 48 4B 48 91 48 D7 49 1D GSG{G^H.HKHatH I.
0000900	49 63 49 A9 49 F0 4A 37 4A 7D 4A C4 4B 0C 4B 53 IcI-I=J7J}J-K.KS
0000910	4B 9A 4B E2 4C 2A 4C 72 4C BA 4D 02 4D 4A 4D 93 KÜKTL*LrL M.MJMô
0000920	4D DC 4E 25 4E 6E 4E B7 4F 00 4F 49 4F 93 4F DD M_N%NnN_O.OIOôO
0000930	50 27 50 71 50 BB 51 06 51 50 51 9B 51 E6 52 31 P'PqP_Q.QPQqQuR1
0000940	52 7C 52 C7 53 13 53 5F 53 AA 53 F6 54 42 54 8F R R S.S_S-S+TBTÂ
0000950	54 DB 55 28 55 75 55 C2 56 0F 56 5C 56 A9 56 F7 T U(UuU+V.V\V-V≈
0000960	57 44 57 92 57 E0 58 2F 58 7D 58 CB 59 1A 59 69 WDWÆWαX/X}X≈Y.Yi
0000970	59 B8 5A 07 5A 56 5A A6 5A F5 5B 45 5B 95 5B E5 YqZ.ZVZ^Z}[E[ò[σ
0000980	5C 35 5C 86 5C D6 5D 27 5D 78 5D C9 5E 1A 5E 6C \5\á_ ' x] ^.^l
0000990	5E BD 5F 0F 5F 61 5F B3 60 05 60 57 60 AA 60 FC ^ _ _a_ ' .'W'-'n
00009A0	61 4F 61 A2 61 F5 62 49 62 9C 62 F0 63 43 63 97 a0aóa bIbfb=cCù
00009B0	63 EB 64 40 64 94 64 E9 65 3D 65 92 65 E7 66 3D còd@dòdee=eÆetf=
00009C0	66 92 66 E8 67 3D 67 93 67 E9 68 3F 68 96 68 EC fÆføg=gôgeh?hùh∞
00009D0	69 43 69 9A 69 F1 6A 48 6A 9F 6A F7 6B 4F 6B A7 iCiÜi±jHj fj≈kOk°
00009E0	6B FF 6C 57 6C AF 6D 08 6D 60 6D B9 6E 12 6E 6B k lWl»m.m'n n.nk
00009F0	6E C4 6F 1E 6F 78 6F D1 70 2B 70 86 70 E0 71 3A n-o.oxo~p+pâpαq:
0000A00	71 95 71 F0 72 4B 72 A6 73 01 73 5D 73 B8 74 14 qòq≡rKr^s.s}sq t.
0000A10	74 70 74 CC 75 28 75 85 75 E1 76 3E 76 9B 76 F8 tpt u(uàußv>vçv°
0000A20	77 56 77 B3 78 11 78 6E 78 CC 79 2A 79 89 79 E7 wVw x.xnx y*yëyτ
0000A30	7A 46 7A A5 7B 04 7B 63 7B C2 7C 21 7C 81 7C E1 zFzN{. {c{t ! ü ß
0000A40	7D 41 7D A1 7E 01 7E 62 7E C2 7F 23 7F 84 7F E5 }A}i~.~b~_~#ôäöσ
0000A50	80 47 80 A8 81 0A 81 6B 81 CD 82 30 82 92 82 F4 ÇGÇz.ü.ükü=é0éKé
0000A60	83 57 83 BA 84 1D 84 80 84 E3 85 47 85 AB 86 0E äwä ä.äçäpäGà½ä.
0000A70	86 72 86 D7 87 3B 87 9F 88 04 88 69 88 CE 89 33 ârâ ç;çfê.êiê ê3
0000A80	89 99 89 FE 8A 64 8A CA 8B 30 8B 96 8B FC 8C 63 ê0ê•edê=î0îüî"îc
0000A90	8C CA 8D 31 8D 98 8D FF 8E 66 8E CE 8F 36 8F 9E î=îiîÿî ÄfÄ Ä6ÄP
0000AA0	90 06 90 6F 90 D6 91 3F 91 A8 92 11 92 7A 92 F3 Ê.FnF≈?æ;Æ.Æ>Æπ

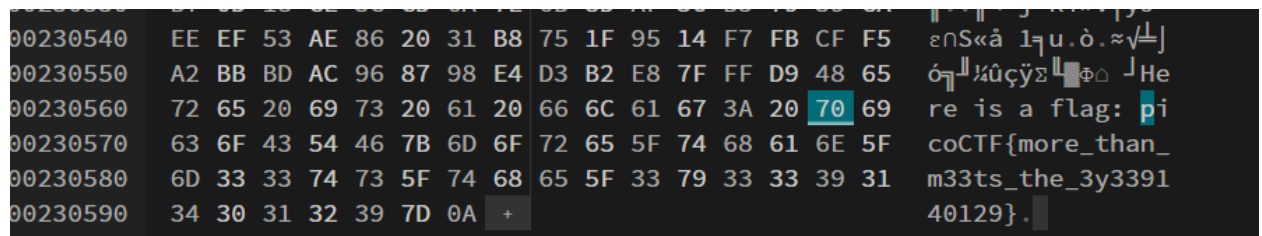


4. Searched for the keyword:

picoCTF



5. Located the flag directly embedded in the file

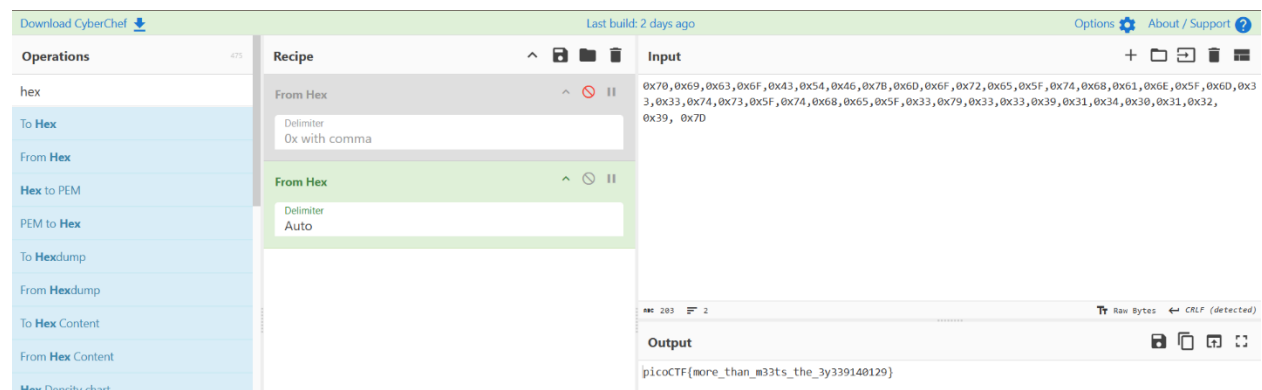


Alternative method:

Copy the hex values

Paste into CyberChef

Use “**From Hex**” to decode

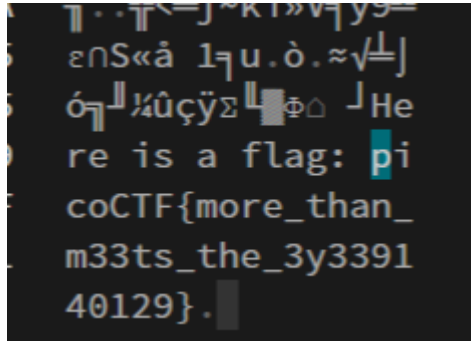




6. Retrieved the **flag**

V. Result

picoCTF{more_than_m33ts_the_3y339140129}



VI. Explanation

How are images made?

A digital image is made of:

- Tiny dots called **pixels**
- Each pixel has a **color value**

But inside the computer...

Everything becomes:

0s and 1s

Why do we use Hexadecimal?

Binary (0s and 1s) is hard to read:

010101010101

So we convert it into **hex**:

AA55



Hex is just a **shorter, easier way to read binary**

What's inside an image file?

An image file contains:

- File header (type info)
- Image data (pixels)
- Sometimes extra hidden data

Think of it like:

A **picture frame** 

- Front = image you see
- Back = hidden notes taped behind

What is a Hex Editor?

A hex editor lets you:

See the **raw contents** of a file

Instead of seeing a picture, you see:

89 50 4E 47 ...

What did we do?

We searched for:

picoCTF

Why? because sometimes:

The flag is stored as plain text inside the file

Why does this work?

Because:

- **Files don't "hide" text perfectly**
- **Sometimes developers leave data inside**
- **The image still works even with extra hidden text**

Real-Life Hacking Example

Hackers can:



- **Hide malicious code inside images**
- **Store secret messages in files**
- **Embed data that normal users can't see**

VII. Conclusion

This challenge demonstrates how digital files contain more information than what is immediately visible.

By analyzing the raw hexadecimal data of the image, hidden text was discovered and the flag was successfully retrieved.

This reinforces a key concept in digital forensics:

Always inspect the raw data — important information may be hidden beneath the surface.



— **The Analyst: Hyposelenia**